

Artificial intelligence (AI) is the capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making. It is a field of research in engineering, mathematics, and computer science that develops and studies methods and software that enable machines to perceive their environment and use learning and intelligence to take actions that maximise their chances of achieving defined goals.

High-profile applications of AI include advanced web search engines, chatbots, virtual assistants, autonomous vehicles, play and analysis in strategy games (e.g., chess and Go), and content generation (e.g. images, audio, and videos).

The traditional goals of AI research include learning, reasoning, knowledge representation, planning, natural language processing, and perception, as well as support for robotics. To reach these goals, AI researchers use techniques including state space search and mathematical optimisation, formal logic, artificial neural networks, and methods based on statistics, operations research, and economics. AI also draws upon psychology, linguistics, philosophy, neuroscience, and other fields. Some companies, such as OpenAI, Google DeepMind, and Meta, aim to create artificial general intelligence (AGI)—AI that can complete nearly any cognitive task at least as well as a human.

Artificial intelligence was founded as an academic discipline in 1956. The field went through multiple cycles of optimism throughout its history, followed by periods of disappointment and loss of funding, known as AI winters. Funding and interest increased substantially after 2012, when graphics processing units (GPUs) started being used to accelerate neural networks, and deep learning outperformed previous AI techniques. This growth accelerated further after 2017 with the transformer architecture. In the 2020s, an AI boom coincided with advances in generative AI, which became widespread and allowed for the creation and modification of media. In addition to AI safety and unintended consequences and harms from the use of AI, ethical concerns, AI's long-term effects, environmental effects, and potential existential risks have prompted discussions of AI regulation.

== Goals ==

The general problem of simulating (or creating) intelligence has been broken down into subproblems. These consist of specific traits or capabilities that researchers expect an intelligent system to display. The traits described below have received the most attention and cover the scope of AI research.

=== Reasoning and problem-solving ===

Early researchers developed algorithms that imitated step-by-step reasoning that humans use when solving puzzles or making logical deductions. By the late 1980s and 1990s, methods were developed for dealing with uncertain or incomplete information, employing concepts from probability and economics. Many of these algorithms were insufficient for solving large reasoning problems because they experienced a "combinatorial explosion", meaning they become exponentially slower as the problems grow. Even humans rarely use the step-by-step deduction that early AI research could model. Humans solve most of their problems using fast, intuitive judgments.

Reasoning models, a type of large language model (LLM) trained to generate intermediate chains-of-thought, emerged in 2024 and allowed improved performance on complex problems in mathematics and coding. These models can produce incorrect outputs or "hallucinations," unlike symbolic reasoning systems.

=== Knowledge representation ===

AI programs use knowledge to answer questions intelligently and make deductions about real-world facts.

Formal knowledge representation and knowledge engineering use symbols to represent words, concepts and things in the world. A knowledge base is a body of knowledge represented in a form that can be used by a program. An ontology is the set of objects, relations, concepts, and properties used by a particular domain of knowledge. Formal knowledge has been studied extensively since the 1970s and researchers have developed formalisms for a wide variety of difficult domains.

The symbolic approach has difficulty with several problems: the breadth of commonsense knowledge (the set of atomic facts the average person knows is enormous), the sub-symbolic form of most commonsense knowledge (much of what people know is not represented as "facts" or "statements" they can express verbally), and knowledge acquisition (the problem of obtaining knowledge for AI applications).

Large language models (and some other AI programs developed since 2012) do not require explicit, symbolic knowledge. They acquire knowledge by being trained on the combined text of millions of books and billions of websites. Modern AI can also learn about a domain by running experiments (as when AlphaZero learns game strategy by playing against itself). Machine learning solves the problems of general knowledge, commonsense knowledge and knowledge acquisition, however this approach has struggled with accurate recall and valid reasoning.

=== Planning and decision-making ===

An "agent" is any entity (artificial or not) that perceives and takes actions in the world. A rational agent has goals or preferences and takes actions to make them happen. In automated planning, the agent has a specific goal. In automated decision-making, the agent has preferences—there are some situations it would prefer to be in, and some situations it is trying to avoid. The decision-making agent assigns a number to each situation (called its "utility") that measures how much the agent prefers it. For each possible action, it can calculate the "expected utility": the utility of all possible outcomes of the action, weighted by the probability that the outcome will occur. It can then choose the action with the maximum expected utility.

In classical planning, the agent knows exactly what the effect of any action will be. In most real-world problems, however, the agent may not be certain about the situation it is in (it is "unknown" or "unobservable") and it may not know for certain what will happen after each possible action (it is not "deterministic"). It must choose an action by making a probabilistic guess and then reassess the situation to see if the action worked.

Alongside thorough testing and improvement based on previous decisions, having an explanation for why the agent took certain decisions is a way to build trust, especially when the decisions have to be relied upon.

In some problems, the agent's preferences may be uncertain, especially if there are other agents or humans involved. These preferences may be learned (e.g., with inverse reinforcement learning), or the agent can seek information to improve them. Information value theory can be used to weigh the value of exploratory or experimental actions. The space of possible future actions and situations is typically intractably large, so the agents must take actions and evaluate situations while being uncertain of the outcome.

A Markov decision process has a transition model that describes the probability that a particular action will change the state in a particular way and a reward function that supplies the utility of each state and the cost of each action. A policy associates a decision with each possible state. The policy could be calculated (e.g., by policy iteration), determined by a heuristic, or learned.

Game theory describes the rational behaviour of multiple interacting agents and is used in AI programs that make decisions involving other agents.

=== Learning ===

Machine learning is the study of programs that can improve their performance on a given task automatically. It has been a part of AI from the beginning.

There are several kinds of machine learning:

Unsupervised learning analyses a stream of data, finds patterns, and makes predictions without any other guidance.

Supervised learning requires labelling the training data with the expected answers, and comes in two main varieties: classification (where the program must learn to predict what category the input belongs in) and regression (where the program must deduce a numeric function based on numeric input). Reinforcement learning is when the agent is rewarded for good responses and punished for bad ones. The agent learns to choose responses that are classified as "good".

Transfer learning is when the knowledge gained from one problem is applied to a new problem.

Deep learning is a type of machine learning that runs inputs through biologically inspired artificial neural networks for all of these types of learning.

Computational learning theory can assess learners by computational complexity, sample complexity (how much data is required), or other notions of optimisation.

=== Natural language processing ===

Natural language processing (NLP) allows programs to read, write, and communicate in human languages. Specific problems include speech recognition, speech synthesis, machine translation, information extraction, information retrieval, and question answering.

Early work, based on Noam Chomsky's generative grammar and semantic networks, had difficulty with word-sense disambiguation unless restricted to small domains called "micro-worlds" (due to the common sense knowledge problem). British linguist and philosopher Margaret Masterman believed it was meaning and not grammar that was the key to understanding languages, and that dictionaries and especially thesauri should be the basis of computational language structure.

Modern deep learning techniques for NLP include word embedding (representing words, typically as vectors encoding their meaning), transformers (a deep learning architecture using an attention mechanism), and others. In 2019, generative pre-trained transformer (or "GPT") language models began to generate coherent text. By 2023, these models were able to get human-level scores on the bar exam, SAT (Scholastic Assessment Test), GRE (Graduate Record Examination), and many other real-world applications.

=== Perception ===

Machine perception is the ability to use input from sensors (such as cameras, microphones, wireless signals, active lidar, sonar, radar, and tactile sensors) to deduce aspects of the world. Computer vision is the ability to analyse visual input.

The field includes speech recognition, image classification, facial recognition, object recognition, object tracking, and robotic perception.

=== Social intelligence ===

Affective computing is a field that comprises systems that recognise, interpret, process, or simulate human affect (feeling, emotion, and mood). For example, some virtual assistants are programmed to speak conversationally or even banter humorously; it makes them appear more sensitive to the emotional dynamics of human interaction, or to otherwise facilitate human-computer interaction.

However, this tends to give naïve users an unrealistic conception of the intelligence of existing computer agents. Moderate successes related to affective computing include textual sentiment analysis and, more recently,

multimodal sentiment analysis, wherein AI classifies the effects displayed by a videotaped subject.

=== General intelligence ===

A machine with artificial general intelligence (AGI) would be able to solve a wide variety of problems with breadth and versatility similar to human intelligence.

== Techniques ==

AI research uses a wide variety of techniques to accomplish the goals above.

=== Search and optimisation ===

There are two different kinds of search used in AI: state space search and local search:

==== State space search ====

State space search searches through a tree of possible states to try to find a goal state. For example, planning algorithms search through trees of goals and subgoals, attempting to find a path to a target goal, a process called means-ends analysis.

Simple exhaustive searches are rarely sufficient for most real-world problems: the search space (the number of places to search) quickly grows to astronomical numbers. The result is a search that is too slow or never completes. "Heuristics" or "rules of thumb" can help prioritise choices that are more likely to reach a goal.

Adversarial search is used for game-playing programs, such as chess or Go. It searches through a tree of possible moves and countermoves, looking for a winning position.

==== Local search ====

Local search uses mathematical optimisation to find a solution to a problem. It begins with some form of guess and refines it incrementally.

Gradient descent is a type of local search that optimises a set of numerical parameters by incrementally adjusting them to minimise a loss function. Variants of gradient descent are commonly used to train neural networks, through the backpropagation algorithm.

Another type of local search is evolutionary computation, which aims to iteratively improve a set of candidate solutions by "mutating" and "recombining" them, selecting only the fittest to survive each generation.

Distributed search processes can coordinate via swarm intelligence algorithms. Two popular swarm algorithms used in search are particle swarm optimisation (inspired by bird flocking) and ant colony optimisation (inspired by ant trails).

=== Logic ===

Formal logic is used for reasoning and knowledge representation.

Formal logic comes in two main forms: propositional logic (which operates on statements that are true or false and uses logical connectives such as "and", "or", "not" and "implies") and predicate logic (which also operates on objects, predicates and relations and uses quantifiers such as "Every X is a Y" and "There are some Xs that are Ys").

Deductive reasoning in logic is the process of proving a new statement (conclusion) from other statements that are given and assumed to be true (the premises). Proofs can be structured as proof trees, in which nodes are labelled by sentences, and children nodes are connected to parent nodes by inference rules.

Given a problem and a set of premises, problem-solving reduces to searching for a proof tree whose root node is labelled by a solution of the problem and whose leaf nodes are labelled by premises or axioms. In the case of Horn clauses, problem-solving search can be performed by reasoning forwards from the premises or backwards from the problem. In the more general case of the clausal form of first-order logic, resolution is a single, axiom-free rule of inference, in which a problem is solved by proving a contradiction from premises that include the negation of the problem to be solved.

Inference in both Horn clause logic and first-order logic is undecidable, and therefore intractable. However, backward reasoning with Horn clauses, which underpins computation in the logic programming language Prolog, is Turing complete. Moreover, its efficiency is competitive with computation in other symbolic programming languages.

Fuzzy logic assigns a "degree of truth" between 0 and 1. It can therefore handle propositions that are vague and partially true.

Non-monotonic logics, including logic programming with negation as failure, are designed to handle default reasoning. Other specialised versions of logic have been developed to describe many complex domains.

=== Probabilistic methods for uncertain reasoning ===

Many problems in AI (including reasoning, planning, learning, perception, and robotics) require the agent to operate with incomplete or uncertain information. AI researchers have devised a number of tools to solve these problems using methods from probability theory and economics. Precise mathematical tools have been developed that analyse how an agent can make choices and plan, using decision theory, decision analysis, and information value theory. These tools include models such as Markov decision processes, dynamic decision networks, game theory and mechanism design.

Bayesian networks are a tool that can be used for reasoning (using the Bayesian inference algorithm), learning (using the expectation-maximisation algorithm), planning (using decision networks) and perception (using dynamic Bayesian networks).

Probabilistic algorithms can also be used for filtering, prediction, smoothing, and finding explanations for streams of data, thus helping perception systems analyse processes that occur over time (e.g., hidden Markov models or Kalman filters).

=== Classifiers and statistical learning methods ===

The simplest AI applications can be divided into two types: classifiers (e.g., "if shiny then diamond"), on one hand, and controllers (e.g., "if diamond then pick up"), on the other hand. Classifiers are functions that use pattern matching to determine the closest match. They can be fine-tuned based on chosen examples using supervised learning. Each pattern (also called an "observation") is labeled with a certain predefined class. All the observations combined with their class labels are known as a data set. When a new observation is received, that observation is classified based on previous experience.

There are many kinds of classifiers in use. The decision tree is the simplest and most widely used symbolic machine learning algorithm. K-nearest neighbour algorithm was the most widely used analogical AI until the mid-1990s, and Kernel methods such as the support vector machine (SVM) displaced k-nearest neighbour in the 1990s.

The naive Bayes classifier is reportedly the "most widely used learner" at Google, due in part to its scalability.

Neural networks are also used as classifiers.

=== Artificial neural networks ===

An artificial neural network is based on a collection of nodes also known as artificial neurons, which loosely model the neurons in a biological brain. It

is trained to recognise patterns; once trained, it can recognise those patterns in fresh data. There is an input, at least one hidden layer of nodes and an output. Each node applies a function and once the weight crosses its specified threshold, the data is transmitted to the next layer. A network is typically called a deep neural network if it has at least 2 hidden layers. Learning algorithms for neural networks use local search to choose the weights that will get the right output for each input during training. The most common training technique is the backpropagation algorithm. Neural networks learn to model complex relationships between inputs and outputs and find patterns in data. In theory, a neural network can learn any function. In feedforward neural networks the signal passes in only one direction. The term perceptron typically refers to a single-layer neural network. In contrast, deep learning uses many layers. Recurrent neural networks (RNNs) feed the output signal back into the input, which allows short-term memories of previous input events. Long short-term memory networks (LSTMs) are recurrent neural networks that better preserve longterm dependencies and are less sensitive to the vanishing gradient problem. Convolutional neural networks (CNNs) use layers of kernels to more efficiently process local patterns. This local processing is especially important in image processing, where the early CNN layers typically identify simple local patterns such as edges and curves, with subsequent layers detecting more complex patterns like textures, and eventually whole objects.

=== Deep learning ===

Deep learning uses several layers of neurons between the network's inputs and outputs. The multiple layers can progressively extract higher-level features from the raw input. For example, in image processing, lower layers may identify edges, while higher layers may identify the concepts relevant to a human such as digits, letters, or faces.

Deep learning has profoundly improved the performance of programs in many important subfields of artificial intelligence, including computer vision, speech recognition, natural language processing, image classification, and others. The reason that deep learning performs so well in so many applications is not known as of 2021. The sudden success of deep learning in 2012–2015 did not occur because of some new discovery or theoretical breakthrough (deep neural networks and backpropagation had been described by many people, as far back as the 1950s) but because of two factors: the increase in computer power (including the hundred-fold increase in speed by switching to GPUs) and the availability of vast amounts of training data, especially the giant curated datasets used for benchmark testing, such as ImageNet.

=== GPT ===

Generative pre-trained transformers (GPTs) are large language models (LLMs) that generate text based on the semantic relationships between words in sentences. Text-based GPT models are pre-trained on a large corpus of text that can be from the Internet. The pretraining consists of predicting the next token (a token being usually a word, subword, or punctuation). Throughout this pretraining, GPT models accumulate knowledge about the world and can then generate human-like text by repeatedly predicting the next token. Typically, a subsequent training phase makes the model more truthful, useful, and harmless, usually with a technique called reinforcement learning from human feedback (RLHF). Current GPT models are prone to generating falsehoods called "hallucinations". These can be reduced with RLHF and quality data, but the problem has been getting worse for reasoning systems. Such systems are used in chatbots, which allow people to ask a question or request a task in simple text.

Current models and services include ChatGPT, Claude, Gemini, Copilot, and Meta AI. Multimodal GPT models can process different types of data (modalities) such as images, videos, sound, and text.

=== Hardware and software ===

In the late 2010s, graphics processing units (GPUs) that were increasingly designed with AI-specific enhancements and used with specialised TensorFlow software had replaced previously used central processing unit (CPUs) as the dominant means for large-scale (commercial and academic) machine learning models' training. Specialised programming languages such as Prolog were used in early AI research, but general-purpose programming languages like Python have become predominant.

The transistor density in integrated circuits has been observed to roughly double every 18 months—a trend known as Moore's law, named after the Intel co-founder Gordon Moore, who first identified it. Improvements in GPUs have been even faster, a trend sometimes called Huang's law, named after Nvidia co-founder and CEO Jensen Huang.

== Applications ==

AI and machine learning technology is used in most of the essential applications of the 2020s, including:

- search engines (such as Google Search)
- targeting online advertisements
- recommendation systems (offered by Netflix, YouTube or Amazon) driving internet traffic
- targeted advertising (AdSense, Facebook)
- virtual assistants (such as Siri or Alexa)
- autonomous vehicles (including drones, ADAS and self-driving cars)
- automatic language translation (Microsoft Translator, Google Translate)
- facial recognition (Apple's FaceID or Facebook's DeepFace and Google's FaceNet)
- image labelling (used by Facebook, Apple's Photos and TikTok).

=== AI-assisted software development ===

=== Chatbots ===

=== Physical AI ===

=== Health and medicine ===

AlphaFold 2 (2021) demonstrated the ability to approximate, in hours rather than months, the 3D structure of a protein. In 2023, it was reported that AI-guided drug discovery helped find a class of antibiotics capable of killing two different types of drug-resistant bacteria. In 2024, researchers used machine learning to accelerate the search for Parkinson's disease drug treatments. Their aim was to identify compounds that block the clumping, or aggregation, of alpha-synuclein (the protein that characterises Parkinson's disease). They were able to speed up the initial screening process ten-fold and reduce the cost by a thousand-fold.

AI is increasingly being used in medical diagnostics, including the detection of diseases such as lung cancer from medical imaging like CT scans.

A 2026 Nature article titled "Dozens of AI disease-prediction models were trained on dubious data" highlighted the use of unreliable data being used to train AI medical prediction models for stroke and diabetes in 125 research articles. Evidence suggested some of the AI tools that were developed on unreliable data had been used on patients, although it was not clear if there were adverse outcomes.

=== Gaming ===

Game playing programs have been used since the 1950s to demonstrate and test AI's most advanced techniques. Deep Blue became the first computer chess-playing system to beat a reigning world chess champion, Garry Kasparov, on 11 May 1997. In 2011, in a Jeopardy! quiz show exhibition match, IBM's question answering system, Watson, defeated the two greatest Jeopardy! champions, Brad Rutter and Ken Jennings, by a significant margin. In March 2016, AlphaGo won 4 out of 5 games of Go in a match with Go champion Lee Sedol, becoming the first computer Go-playing system to beat a professional Go player without handicaps. Then, in 2017, it defeated Ke Jie, who was the best Go player in the world. Other programs handle imperfect-information games, such as the poker-playing program Pluribus. DeepMind developed increasingly generalistic reinforcement learning models, such as with MuZero, which could be trained to play chess, Go, or Atari games. In 2019, DeepMind's AlphaStar achieved grandmaster level in StarCraft II, a particularly challenging real-time strategy game that involves incomplete knowledge of what happens on the map. In 2021, an AI agent competed in a PlayStation Gran Turismo competition, winning against four of the world's best Gran Turismo drivers using deep reinforcement learning. In 2024, Google DeepMind introduced SIMA, a type of AI capable of autonomously playing nine previously unseen open-world video games by observing screen output, as well as executing short, specific tasks in response to natural language instructions.

=== Mathematics ===

In mathematics, probabilistic large language models are versatile, but can also produce wrong answers in the form of hallucinations. The Alibaba Group developed a version of its Qwen models called Qwen2-Math, that achieved state-of-the-art performance on several mathematical benchmarks, including 84% accuracy on the MATH dataset of competition mathematics problems. In January 2025, Microsoft proposed the technique rStar-Math that leverages Monte Carlo tree search and step-by-step reasoning, enabling a relatively small language model like Qwen-7B to solve 53% of the AIME 2024 and 90% of the MATH benchmark problems. Google DeepMind has developed models for solving mathematical problems: AlphaTensor, AlphaGeometry, AlphaProof, AlphaEvolve, and FunSearch.

When natural language is used to describe mathematical problems, converters can transform such prompts into a formal language such as Lean to define mathematical tasks. The experimental model Gemini Deep Think accepts natural language prompts directly and achieved gold medal results in the International Math Olympiad of 2025.

Topological deep learning integrates various topological approaches.

=== Finance ===